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Digital Modulation Techniques Simulation Using MATLAB

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Different types of modulation techniques are used in telecommunication for transmitting and receiving radio frequency (RF) signals. With MATLAB tools we can analyse the modulation techniques used for these signals along with their mathematical representation. We can analyse carrier frequency parameters like amplitude, phase, and the frequency. Simulation with MATLAB tools helps us obtain the real-world characteristics.

Digital modulation is digital-to-analogue conversion (DAC) while demodulation or detection is analogue-to-digital conversion (ADC). The modulated digital signal can be transmitted through various media like cables, microwaves, or satellites. The following types of digital modulation techniques are simulated in this project:

1. Phase shift keying (PSK): (a) binary phase shift keying (BPSK), (b) quadrature phase shift keying (QPSK), and (c) eight phase shift keying (8PSK)
2. Amplitude shift keying (ASK)
3. On-off keying (OOK)
4. Quadrature amplitude modulation (QAM)

BPSK modulation

In PSK modulation, phase of carrier signal is changed in accordance with the input digital data. It is a digital modulation technique that is used for video, aircraft, and satellite signals. The most straightforward type of PSK is called Binary Phase Shift Keying (BPSK), where 'binary' refers to the use of two phase offsets (one for logic high, one for logic low). That is, BPSK is a two-phase modulation scheme, where the 0's and 1's in a binary message are represented by two different phase states in the carrier signal: $\theta = 0°$ for binary 1 and $\theta = 180°$ for binary 0.

We can think of BPSK as simply inverting the carrier in response to one logic state and leaving it alone in response to the other logic state.

In BPSK modulation, the phase of the RF carrier is shifted 180 degrees in accordance with a digital bit stream (modulating input binary signal). A '1' causes a phase transition, and a '0' does not produce any transition. However, both peak amplitudes of the BPSK signals remain constant as phase changes.

Run the *digital_modulation.m* file and select the BPSK option from the drop-down menu on the right side of the screen. You will get the simulated output of BPSK modulated signal as shown in Fig. 1.

The incoming modulated BPSK signal is multiplied with carrier signal generated by the carrier generator. The output of the multiplier contains high-frequency and low-frequency components.

QPSK modulation

QPSK is a form of PSK in which two

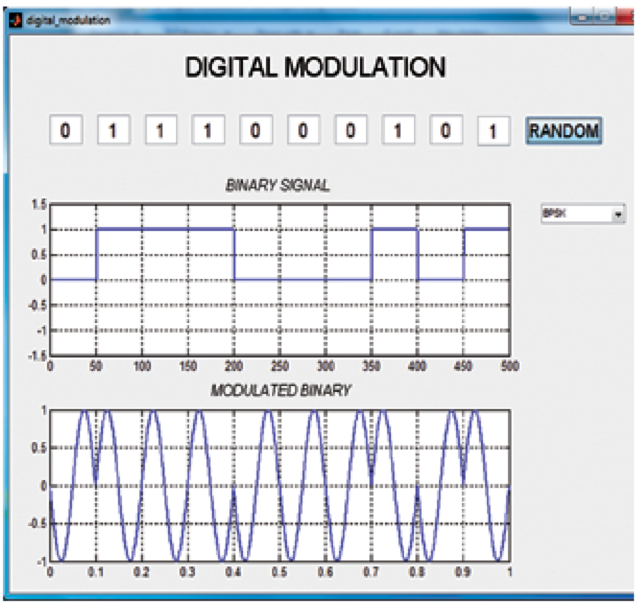

Fig. 1: BPSK modulation

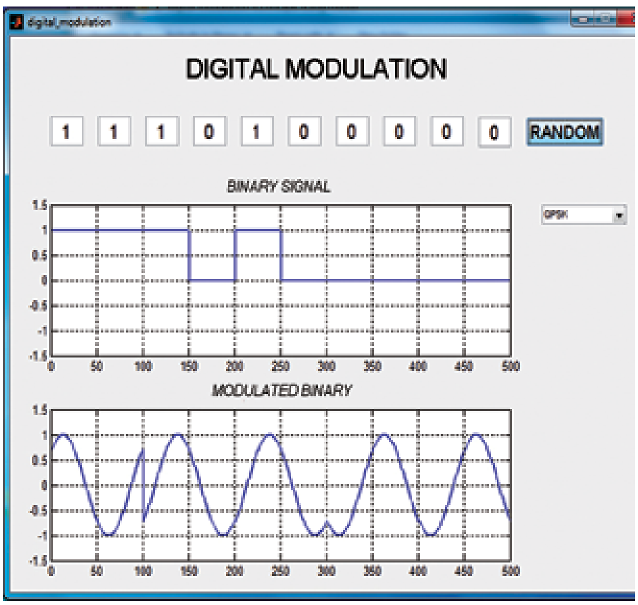

Fig. 2: QPSK modulation